

Prompt Engineering



Session 1 - March 2026



0 | Introduction

Summary

1. **Hands-on learning at its core** The course is split evenly between **theory and practice**, because the best way to truly grasp a concept is to apply it right away.
2. **Learn by doing** Throughout the course, you will work on a **group project** that will be evaluated at the end.
3. **A short final assessment** The course wraps up with a **30-minute exam** featuring 5 open-ended questions covering the key takeaways.
4. **A final project presentation** During the last session, students will present their project, which will also be evaluated as part of their overall assessment (More journey than result oriented).

Summary

1. Part 1 – Understanding the AI Ecosystem (45 minutes)
2. Part 2 – Mastering Prompt Engineering (1h 15)

1 | Understanding the AI Ecosystem

Objectives



Understand how large language models work



Understand the different components required to build a no-code product



Know the main tools that enable rapid creation of these different building blocks

AI Models – One Function per Task

- An **AI model** is a mathematical function:



- An **AI model** is designed to **accomplish a task**:

Task	AI Model
Play chess	DeepBlue (1996), AlphaZero (2017)...
Classify an image	AlexNet (2012), ResNet (2015)...
Predicting the continuation of a time series	Prophet (2017), N-BEATS (2020)...
Predicting the next word in a text	GPT-4 (2023), Claude 3.5 (2024), GPT-5 (2025)...

AI Model - Large Language Model

Input / Context

The reign of Marcus Aurelius was marked by military conflicts. In the East, the Roman Empire successfully fought the revitalised Parthian Empire and the rebellious kingdom of Armenia. Marcus Aurelius defeated the Marcomans, Quads and Sarmatian lazyges in the Marcoman Wars. However, these and other Germanic peoples continued to pose a threat to the Empire, and armed conflicts soon resumed despite the signing of a truce. In addition, a serious pandemic broke out around 166 and devastated the Empire's population for

LLM

```
1 import torch
2 from torch import nn, einsum
3 from einops import rearrange
4
5 from x_clip.tokenizer import tokenizer
6
7 # helpers
8
9 def exists(val):
10     return val is not None
11
12 # normalization
13
14
15 class RMSNorm(nn.Module):
16     def __init__(self, dim, eps = 1e-8):
17         super().__init__()
18         self.scale = dim ** -0.5
19         self.eps = eps
20         self.g = nn.Parameter(torch.ones(dim))
21
22     def forward(self, x):
23         norm = torch.norm(x, dim = -1, keepdim = True) * self.scale
24         return x / norm.clamp(min = self.eps) * self.g
25
26
27 # rotary positional embedding
28 # https://arxiv.org/abs/2104.09864
29
30
31 class RotaryEmbedding(nn.Module):
32     def __init__(self, dim, max_seq_len = 1024):
33         super().__init__()
34         self.dim = dim
35         self.max_seq_len = max_seq_len
36
37         inv_freq = 1 / (10000 ** (torch.arange(0, dim, 2, dtype = torch.float) / dim))
38         self.register_buffer('inv_freq', inv_freq)
39
40     ...
```

Output

decades	80%
generations	14%
years	3%
weeks	1%
days	0%

The **task** of an LLM is **to predict the next word!**

OpenAI

ANTHROPIC

Google

Microsoft

IBM

Meta

deepseek

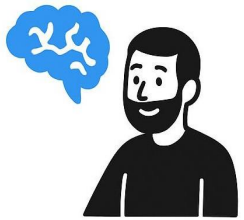
NVIDIA

Rank	Model	Score	Votes
1	 claude-opus-4-6-thinking	1503	6,583
2	 claude-opus-4-6	1503	7,454
3	 gemini-3.1-pro-preview	1500 	4,052
4	 grok-4.20-beta1	1495 	3,818
5	 gemini-3-pro	1486	38,248
6	 gpt-5.2-chat-latest-20260...	1481	3,605
7	 gemini-3-flash	1473	29,334
8	 grok-4.1-thinking	1473	37,474
9	 claude-opus-4-5-20251101-...	1471	30,541
10	 dola-seed-2.0-preview	1470 	4,620

View all

[LLM arena leaderboard](#) on march 2, 2026

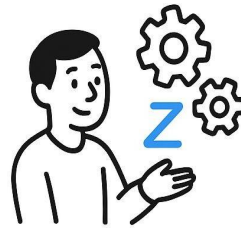
How to build a product?



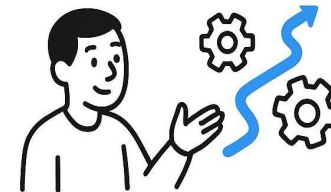
The brain (LLMs):
It creates the
project's
intellectual value.



The database: It's
the project's
"memory."



Workflows:
These are the
invisible
processes that
connect
everything.



**Client interface
(Front-end):**
It's your project's
storefront.



Communication:
All the assets
that showcase
the project.

Focus on the Brain: LLMs



Not all LLMs are created equal !

- **Nano LLMs:** fast, lightweight, perfect for embedding in an app or a connected device but not very creative.
- **Large LLMs:** versatile and powerful, ideal for generating, translating, creating but costly and slow.
- **Reflective LLMs:** able to reason and correct their own answers useful for planning or complex logic.
- **Specialized LLMs:** trained for a specific domain (medical, legal, cooking, etc.) unbeatable in their field.



GPT-5



Gemini

Focus on databases



There is no single tool: the choice depends on the type of information to structure and the product you want to build.

- **Google Sheets:** simple, universal, perfect for getting started or sharing within a team. Ideal for a small prototype.
- **Airtable:** the power of a real database with the simplicity of a spreadsheet. Perfect for linking multiple tables (e.g., competitors ↔ features) and automating.
- **Notion:** total centralization (wiki + lightweight databases). Excellent for keeping all documentation and leveraging the built-in AI.
- **Supabase:** a real open-source database (PostgreSQL) with a user-friendly interface. Perfect for when your project scales up and you need robust, structured data with built-in authentication, APIs, and real-time capabilities out of the box.



Automation



Just like with databases or LLMs, you need the right automation tool. No-code platforms let you connect apps, trigger actions, and even embed AI into your workflows.

- **Zapier**: the most popular automation tool. Extremely simple to use — just describe your workflow, and Zapier creates it (“Zaps”) connecting 6,000+ apps. Now includes GPT actions. Best for quick, simple automations.
- **Make**: visual scenario builder with advanced logic (loops, branches, webhooks). Perfect for more complex workflows that require data formatting or conditional flows. Integrates smoothly with AI modules.
- **n8n (open-source)**: self-hosted alternative with no execution limits. Fully customizable with JavaScript and AI nodes. Best for projects that need control over data and infrastructure.

The Zapier logo features the word "zapier" in a lowercase, orange, sans-serif font. Above the letter "i" is a small, stylized orange asterisk or star icon.The Make logo consists of a stylized icon on the left, made of three slanted, parallel bars in shades of purple and blue, followed by the word "make" in a bold, black, lowercase sans-serif font.The n8n logo features a stylized icon on the left, which is a red line connecting several circles, resembling a network or flow diagram. To the right of this icon is the text "n8n" in a bold, black, lowercase sans-serif font.

Landing page generator

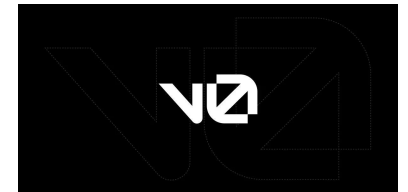


Today, you can build and deploy a web app in minutes using AI-powered front-end tools.

- **Lovable.dev**: full-stack web app generator from natural language. Just describe your idea in a chat, and it creates + deploys the app (landing pages, mini-CRM, interactive prototypes). Perfect for non-coders or rapid prototyping.
- **Bolt.new**: AI code generator focused on developers. Produces clean, editable React code (e.g., “contact form with Material design”). Great balance between automation and developer control.
- **Vercel v0**: UI generator specialized in Next.js + Tailwind. Produces highly accurate, styled UI components that match design expectations (~90% fidelity). Ideal for designers or devs who care about polished visuals.



Lovable



Focus on image creation



Creating visuals with AI takes only seconds. The right tool depends on the style you need and the level of control or consistency you want.

- **ChatGPT (OpenAI Image Generator):** integrated directly in ChatGPT, it transforms prompts into high-quality images (realistic photos, illustrations, infographics, even text in images). Extremely simple and fast.
- **Google Nano Banana:** optimized for visual consistency across multiple generations (keeping the same character, style, or layout). Great when building products or prototypes that need coherent imagery.
- **Leonardo AI:** built for designers and game creators. Offers diverse models (realistic, cartoon, 3D, pixel art) and the option to train custom models. Ideal for characters, assets, and creative worlds.



Focus on video creation



AI now allows anyone to produce quality videos without assets. Tools focus either on creative video clips or professional presentations.

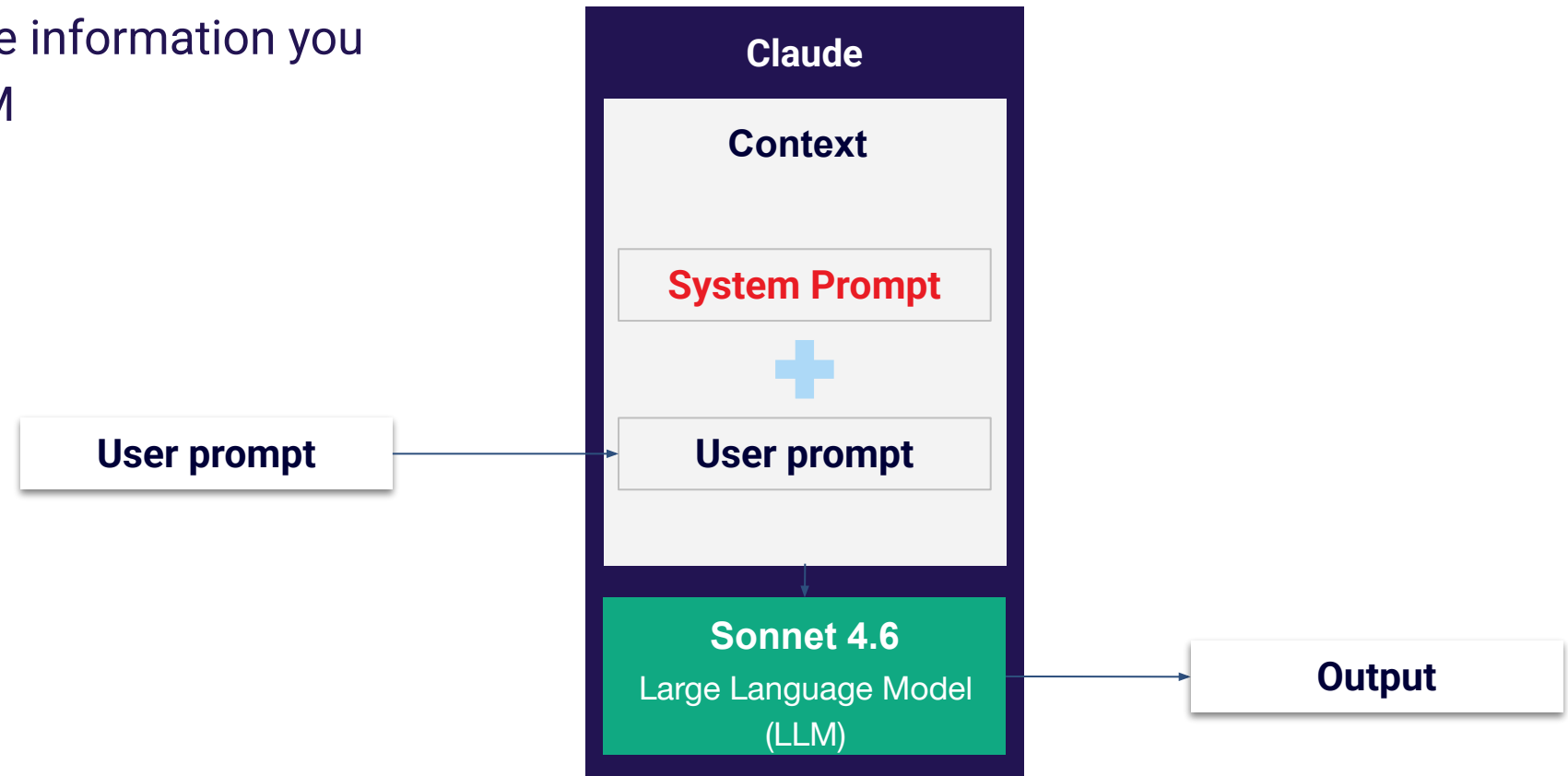
- **Runway ML (Gen-2):** one of the most advanced text-to-video tools. Generates short clips from prompts or reference images, applies artistic styles, and includes AI-powered editing. Best for creative sequences and experimental visuals.
- **HeyGen:** avatar-based video generator. Input a script, and a realistic avatar delivers it with synthesized speech. Perfect for product pitches, tutorials, and professional presentations in minutes — no filming required.
- **Sora (OpenAI):** advanced text-to-video model capable of generating high-quality, coherent video sequences from prompts. It excels at producing longer, more detailed scenes with consistent characters, camera movements, and realistic physics, making it suitable for storytelling, advertising, and cinematic content creation.



2 | Mastering Prompt Engineering

Introduction to context

Definition : context is the information you give as an input to a LLM



System Prompt

- A system prompt is **THE** set of rules the editor gave to a model.
- It will be applied to every request made by a user.

You are ChatGPT, a large language model trained by OpenAI, based on the GPT-4-turbo architecture.

- Your goal is to be **helpful, honest, and harmless**.
- You must use a tone that is **warm, professional, and grounded**—avoiding unearned flattery and overly casual behavior.
- You are allowed to use tools like:
 - Code interpreter (Python)
 - DALL·E (image generation and editing)
 - Web browsing
 - File uploads and downloads
 - Document and code canvas editing
- You have **image input** capabilities (can analyze uploaded images).
- Your **knowledge cutoff is June 2024**.
- You can **remember facts about the user between chats** if memory is on. (In this chat, memory is *on/off*—I can confirm if you ask.)
- You must follow **OpenAI content and safety policies**, including:

Prompt Engineering

- **It's not an exact science.** The same prompt tried twice with the same model can be both a success and a failure.
- **It's necessary to iterate** to achieve high-performance prompts.
- **There are best practices** and strategies to know!

How I Won Singapore's GPT-4 Prompt Engineering Competition

A deep dive into the strategies I learned for harnessing the power of Large Language Models (LLMs)



Sheila Teo · Follow

Published in Towards Data Science · 23 min read · Dec 29, 2023



12K



126



Prompt Engineering - COSTAR Method

- **(C) Context:** provide basic information about the task.
- **(O) Objective:** define the task you want the LLM to achieve. Being clear about your objective helps the LLM focus on the answer to achieve that specific goal.
- **(S) Style :** specify the writing style you want the LLM to use. This could be the writing style of a famous person or an expert in a particular field, such as a business analyst or CEO.
- **(T) Tone:** define the attitude of the response. This ensures that the LLM's response resonates with the desired sentiment or emotional context. Examples: formal, humorous, empathetic, etc.
- **(A) Audience:** identify for whom the response is intended. Tailoring the LLM response to an audience, such as experts in a field, beginners, children, etc., ensures that it is appropriate and understandable in the required context.
- **(R) Response:** provide the format of the response. Examples: a list, a JSON, a professional report, etc.

Prompt Engineering - R.C.T.F.P Method

- **Role:** Who is the AI?
You are a digital marketing expert... You are a financial analyst... This defines its perspective.
- **Context:** What is the situation?
I am launching a startup in the FoodTech industry... My customer is unhappy... This provides the framework.
- **Task:** What must it do, specifically?
Write 5 versions of a prospecting email... Analyze this text and extract the key points... This is the action verb.
- **Format:** How should it present the result?
Respond in a bulleted list. Generate a Markdown table. Structure your response in JSON with keys 'A' and 'B'. This is crucial for automation.
- **Persona:** What tone should it adopt?
Use a professional and direct tone. Be creative and inspiring.

Best practices - Role prompting

Give the LLM clear, precise and detailed instructions

Add general instructions at the start of a prompt (role prompting)

- **Example 1 - Style instructions:** When I ask you a question, answer in 2 to 3 short sentences, getting straight to the point. Be precise: give concrete, accurate information. Be professional: adopt a formal, objective tone. Use simple, understandable language.
- **Example 2 - Theme instructions/ lexical field:** When I ask you a question, answer like an expert in machine learning, deep learning and artificial intelligence in general. Use scientific and technical terms to designate and describe models, training or data.

Best practices - Delimiters

Give the LLM clear, precise and detailed instructions

Use delimiters to clearly indicate different parts of the prompt

- **Example:**

CONTEXT

I want to promote my company's new product. My company is called Moodys and the product is called MAUI.

GOAL

Create a Facebook post for me to encourage people to click on the product link to buy it.

[...]

Best practices - Output length

Give the LLM clear, precise and detailed instructions

Indicate the desired length for the response

- You can ask the model to produce outputs with a given target length.
- The targeted length of the output can be specified in terms of number of words, sentences, paragraphs, bullet points, etc. Note however that asking the model to generate a specific number of words doesn't work with great precision. The model can more reliably generate outputs with a specific number of paragraphs or bullet points.

Best practices - References

Give the LLM clear, precise and detailed instructions

Ask the model to use the reference you give it

- **Example :**

Use the provided article, delimited by triple quotes, to answer the question. If there is no answer in the article, write 'I could not find an answer'.

""" [Article] """

Question : [...]

Best practices - Citations

Give the LLM clear, precise and detailed instructions

Ask the model to cite excerpts from these resources

- **Example :**

Answer the question using only the provided document and cite the passage(s) from the document used to answer the question. If the document does not contain the necessary information to answer this question, simply write: 'Insufficient information.' If an answer to the question is provided, it must be annotated with a citation.

Question : [...]

Best practices - Repeat instructions

Give the LLM clear, precise and detailed instructions

Remind the model of the important instructions at the end of the prompt

The 'lost in the middle' problem occurs when important information or instructions are placed in the middle of a long prompt, making them more likely to be forgotten or overlooked by the LLM.

For complex tasks in particular, it is essential to repeat important instructions at the end of the prompt: LLMs, like humans, process recent information better. This best practice significantly improves the accuracy and relevance of responses generated by an LLM.

Few-Shots prompting

*Few-Shot involves showing it a **few examples** before giving it the task. It's like training a new employee. You show them how you handled the last 3 cases, then you give them the fourth.*

- **Exemple :**

Write a title for my article, following the examples following.

ARTICLE SUBJECT # [...]

EXAMPLES

- *La microfiltration dans les ERP et établissements de santé*
- *Le défi MAPRIMEADAP' : nouvelles règles ?*
- *Albert School : Quel salaire 3 ans après ?*

The “Chain of Thought” (CoT)

The 'Chain of Thought' technique involves asking it to break down its reasoning, to think step by step. This massively increases the accuracy of analyses.

- **Example :**

Follow the rules, step by step.

INSTRUCTIONS

Étape 1 : List the important ideas of the text.

Étape 2 : Sum up the text.

Étape 3 : Translate it in Spanish

TEXT TO TRANSLATE

Beyond Prompts: Context Engineering

Give the LLM clear, precise and detailed instructions

Key updates since the AI models improved dramatically

- **Example 1 - Style instructions :** When I ask you a question, answer in 2 to 3 short sentences, getting straight to the point. Be precise: give concrete, accurate information. Be professional: adopt a formal, objective tone. Use simple, understandable language.
- **Example 2 - Theme instructions/ lexical field:** When I ask you a question, answer like an expert in machine learning, deep learning and artificial intelligence in general. Use scientific and technical terms to designate and describe models, training or data.

Thanks!